

Engineering Executive and 2x Founder with 20+ years of experience who cut production downtime by 70%, raised \$4M, and shipped AI systems used by 480M+ combined DAU across Snap, Amazon, and Google. Expert in bridging R&D innovation with high-security, mission-critical platforms (SOC 2, HIPAA, NIST). Author of 30+ patents/publications with 2,200+ citations.

CORE COMPETENCIES

- ▶ **AI/ML:** Agentic AI, LLM Orchestration, Computer Vision, NLP, Recommendation Systems, Semi-supervised Learning.
- ▶ **Compliance:** SOC 2 Type II, HIPAA, Penetration Testing, Vulnerability Remediation, Data Privacy.
- ▶ **Infrastructure:** Distributed Systems, Platform Architecture, Real-time Processing (100fps), .NET 8, Node.js, Express.js, Sequelize.js, AWS, CI/CD Pipelines.
- ▶ **Leadership:** Team Scaling (2-50+), M&A Technical Due Diligence, Product-Market Fit Strategy, \$2M+ Budget Management.
- ▶ **Languages:** JavaScript and C/C++ expert.

WORK EXPERIENCE

— EXECUTIVE & LEADERSHIP —

Founder / General Partner

March 2023 – Present

Crossroads Venture Studio, Inc.

Direct technical architecture and product-market fit strategy across a portfolio of AI-driven ventures.

- ▶ **Venture Incubation:** Orchestrated technical validation and go-to-market execution for 20+ internal and external teams.
- ▶ **Commercial Performance:** Achieved a 61.5% success rate for portfolio ventures reaching seed funding or first-customer bootstrapping.
- ▶ **Architectural Standardization:** Built a proprietary incubation framework to accelerate AI product delivery, reducing time-to-market for generative AI features and automated CI/CD pipelines.
- ▶ **Venture Strategy:** Serve as primary technical architect for portfolio companies, defining AI orchestration and infrastructure strategies.

VP, Engineering

January 2025 – August 2025

iScribeHealth

- ▶ **Leadership:** Managed a 10-person engineering organization, driving cross-functional alignment across product, engineering, and customer success teams while managing a \$2M annual budget.
- ▶ **Strategic Alignment:** Partnered with CEO and Product stakeholders to align technical roadmap and OKRs with a \$100M ARR growth target for AI-driven clinical documentation.

- ▶ **Security & Compliance:** Guided the organization to SOC 2 and HIPAA compliance; oversaw remediation of 100% of high-severity vulnerabilities from the company's first-ever penetration test.
- ▶ **Infrastructure Modernization:** Drove platform migration from .NET 4 to .NET 8, resulting in a 70% reduction in production downtime.
- ▶ **AI Product Delivery:** Drove delivery of AI-driven E&M coding suggestions, improving provider billing outcomes by 10–20%. Expanded integration with Athena's ambient scribe system for 200+ customers.
- ▶ **Operational Velocity:** Oversaw authentication migration from Azure Entra ID to Auth0; recruited Lead Engineer and PM to strengthen cross-functional execution.
- ▶ **Accountability:** Owned engineering delivery, platform reliability, and production readiness across all customer-facing systems.

Co-Founder, Chief Science Officer & Head of Product

May 2019 – March 2023

Heali AI

Co-founded an AI-driven HealthTech startup focused on personalized nutrition for chronic disease management. Shaped product vision and translated complex nutritional science into a scalable B2B/B2C platform.

- ▶ **Venture Success:** Secured \$4M in seed funding and reached an 8-digit business valuation by establishing the core intellectual property and technical roadmap.
- ▶ **AI & Data Engineering:** Architected a semi-supervised AI pipeline to aggregate and clean 2B+ nutrition data points, building a proprietary engine supporting 200+ medical conditions.
- ▶ **Product & Growth:** Initiated discovery, established prioritization framework, and launched B2C applications on iOS and Android, scaling to 10K+ active users.
- ▶ **Strategic Partnerships:** Bootstrapped a CDC-compliant Diabetes Prevention Program (DPP) product, securing two nationwide healthcare provider partnerships.
- ▶ **Organizational Leadership:** Recruited and directed a multidisciplinary organization of up to 12 engineers, product designers, and UX researchers.
- ▶ **Scientific Contribution:** Authored 1 patent and 1 peer-reviewed publication directly linked to the platform's core AI innovations.

— R&D PRODUCT & ENGINEERING IN BIG TECH INDUSTRY —

Technical Program Manager / Senior Research Scientist

May 2016 – May 2019

Snap Inc.

Directed a 50+ person R&D organization across 40+ concurrent projects, shipping research into global products for 188M+ DAU.

- ▶ **R&D Strategy & Leadership:** Expanded into 10+ research domains, including Face Analysis, NLP, Scene Understanding, Augmented Reality (AR), and Video Analysis.
- ▶ **Product Impact & Revenue:** Converted research into shipped products with measurable business outcomes: Privacy-Preserving Friend Recommendations (14% revival of inactive users, 7% daily engagement); Bitmoji Chat Recommendations (8% increase in user engagement); Sentiment-Based Ticket Sorting (71% increase in support productivity).
- ▶ **Cross-Functional Orchestration:** Coordinated stakeholder management across 5 departments (Business, Design, Data Science, Engineering, Legal) to align R&D roadmaps with global product strategy.

- ▶ **M&A and Intellectual Property:** Served as a core member of the Technical Acquisition Committee; authored 8 patents and 7 publications.

Machine Learning Researcher / Technical Product Manager

November 2012 – May 2016

Amazon · A9

Designed and delivered high-scale computer vision products for A9's visual search team, servicing 200M+ DAU across the global Amazon ecosystem.

- ▶ **Product Innovation:** Launched the mobile brand logo detection and recognition project, achieving real-time performance at 100fps on mobile hardware.
- ▶ **Infrastructure Architecture:** Blueprinted the global visual search database and the proprietary annotation/evaluation platform, managing a corpus of 12M+ images and labels.
- ▶ **IP Development:** Authored and was granted 9 patents for innovations in visual search, object recognition, and machine learning infrastructure.
- ▶ **Strategic Roadmap:** Owned full product lifecycle from R&D through production, ensuring technical feasibility matched consumer requirements.

Machine Learning Engineer

July 2011 – November 2012

Google · YouTube

Initiated and supervised the "Suggested Edits" project, a self-correcting system for large-scale video quality detection and correction.

- ▶ **Global Scale Execution:** Engineered and scaled the feature across the entire YouTube video corpus, impacting 94M+ DAU and improving global video playback quality.
- ▶ **Feature Innovation:** Created 10+ distinct video effects and filters, integrating complex computer vision algorithms into user-facing creator tools.
- ▶ **Intellectual Property:** Secured 2 patents for novel video processing and enhancement techniques.

Application Engineer

June 2003 – June 2004

Infineon Technologies

- ▶ **Systems Programming:** Engineered test platforms and optimized hardware drivers in C and Assembly, reducing validation cycle time by 30%.
- ▶ **Automation:** Built TCL scripts for electronic board test automation, cutting manual testing effort by 40%.

EDUCATION & RESEARCH

- ▶ **PhD** in Computer Science and Artificial Intelligence, Oregon State University, Corvallis, OR June 2011
- ▶ Master in Electrical Engineering and Computer Science, CPE Lyon, France 2005
- ▶ 30+ patents and publications with 2,200+ citations: scholar.google.com/citations?user=mR3fVeIAAAAJ
- ▶ **IEEE/CVF CVPR** — Publication Chair — 2,000+ papers/edition 2017–2019
- ▶ **IEEE WACV** — Publication Chair — 200+ papers 2018–2019

SKILLS

AI/Machine Learning, AI Agent, LLM Orchestration, Computer Vision, NLP, RAG, Distributed Systems, AWS, Node.js, Express.js, Sequelize.js, GitHub Automations, .NET 8, SOC 2, HIPAA, Incident Response, SLAs, Team Scaling, M&A Due Diligence, Budget Management, P&L, Hiring, Product Strategy, Agile, Roadmap, CI/CD, Jira, Asana, Trello, Figma